

HW5

STAT 651

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1

Apply a normal approx. to the posterior of the Zero-Inflated Poisson. Plot the contour of the normal approx, with a scatterplot laid on top

```
suppressPackageStartupMessages(library("MASS"))

# Data
y <- c(rep(0, 25), rep(1, 6), rep(2, 4), rep(3, 3), 5)

log_lik <- function(p, lambda, x) {
  # Log-likelihood components
  loglik_zero <- log(p + (1 - p) * exp(-lambda))
  loglik_nonzero <- log(1 - p) - lambda + x * log(lambda) - lgamma(x + 1)

  # Combine
  sum(ifelse(x == 0, loglik_zero, loglik_nonzero))
}

logp_unnorm <- function(p, l, y) {
  log_lik(p, l, y) + dbeta(p, 1, 1, log = TRUE) + dgamma(l, 2, 1, log = TRUE)
}

res <- optim(c(0.0, 1.0), \(par) -logp_unnorm(p = par[1], l = par[2], y=y), hessian = TRUE)

appx_mean <- res$par
appx_sigma <- solve(res$hessian)

sim <- mvrnorm(n = 10e3, mu = appx_mean, Sigma = appx_sigma)
dens <- kde2d(x = sim[, 1], y = sim[, 2])
# Plotted below

### Draws from posterior using Gibbs (same code from HW4, for comparison)
logfc_pi <- function(p, lambda, x) {
```

```

    log_lik(p, lambda, x) + dbeta(p, 1, 1, log = TRUE)
  }

logfc_lambda <- function(p, lambda, x) {
  log_lik(p, lambda, x) + dgamma(lambda, 2, 1, log = TRUE)
}

# Initialize things
n_pi <- 200
pi_seq <- seq(0, 1, length = n_pi)
n_lambda <- 500
lambda_seq <- seq(0, 30, length = n_lambda)

state <- list(pi = .5, lambda = 1.5)
n_iter <- 1000
sims <- as.data.frame(matrix(NA, nrow = n_iter, ncol = length(state)))
colnames(sims) <- names(state)

for (i in 1:n_iter) {
  ## Update Pi
  lp_pi <- sapply(pi_seq, \(p) logfc_pi(p = p, lambda = state$lambda, x = y))
  state$pi <- sample(pi_seq, 1, prob = exp(lp_pi - max(lp_pi))) # It auto-standardizes the prob

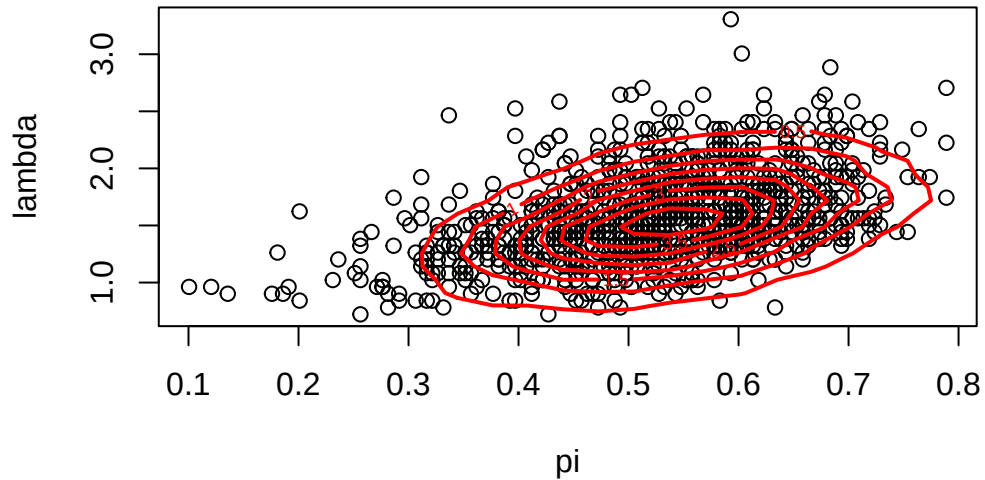
  ## Update Lambda
  lp_lambda <- sapply(lambda_seq, \(l) logfc_lambda(p = state$pi, lambda = l, x = y))
  state$lambda <- sample(lambda_seq, 1, prob = exp(lp_lambda - max(lp_lambda))) # It auto-stand

  sims[i, 'pi'] <- state$pi
  sims[i, 'lambda'] <- state$lambda
}

# Plot the sampled posterior points
plot(sims$pi, sims$lambda, xlab = "pi", ylab = "lambda", main = 'Contour of Norm. Approx. vs. I
contour(dens$x, dens$y, dens$z, col = "red", lwd = 2, add = TRUE)

```

Contour of Norm. Approx. vs. Plot of Sampled Post.



The normal approximation appears to be pretty good within the high-density areas: however, I have concerns that it would not capture the ‘tail’ behaviors of the sampled posterior very well (such as the bottom-left or top-middle of the scatterplot).

2

$$f(y_i|\eta) = h(y_i)$$

Conjugate prior (n_0, y_0 are known):

$$\pi(\eta) \propto e^{-n_0 A(\eta) + n_0 y_0 \eta}$$

a)

$$\begin{aligned} \pi(\eta|y) &\propto L(\eta|y_i)\pi(\eta) \\ &\propto \left[\prod e^{-A(\eta) + y_i \eta} \right] e^{-n_0 A(\eta) + n_0 y_0 \eta} \\ &= e^{-n A(\eta) + \sum y_i \eta} e^{-n_0 A(\eta) + n_0 y_0 \eta} \\ &= e^{-(n+n_0)A(\eta) + (n\bar{y} + n_0 y_0)\eta} \end{aligned}$$

$$\hat{\eta} \text{ occurs when } \frac{d}{d\eta} \pi_{post} = 0$$

For all the following, just assume on all my derivatives I have the $|_{\eta=\hat{\eta}}$. I just don’t feel like repeating that for every step.

$$\begin{aligned}
\frac{d}{d\eta}\pi(\eta|y) &= \frac{d}{d\eta}e^{-(n+n_0)A(\eta)+(n\bar{y}+n_0y_0)\eta} \\
&= \left[\frac{d}{d\eta} - (n+n_0)A(\eta) + (n\bar{y}+n_0y_0)\eta \right] e^{-(n+n_0)A(\eta)+(n\bar{y}+n_0y_0)\eta} \\
&= \left[-(n+n_0)\left[\frac{d}{d\eta}A(\eta)\right] + (n\bar{y}+n_0y_0) \right] e^{-(n+n_0)A(\eta)+(n\bar{y}+n_0y_0)\eta} = 0
\end{aligned}$$

The exponential portion can never equal zero, so we need the coefficient to equal 0:

$$\begin{aligned}
-(n+n_0)\left[\frac{d}{d\eta}A(\eta)\right] + (n\bar{y}+n_0y_0) &= 0 \\
\implies \frac{d}{d\eta}A(\eta) &= \frac{n\bar{y}+n_0y_0}{n+n_0} \\
\implies \frac{d}{d\eta}A(\eta) &= \frac{n_0}{n_0+n}y_0 + \frac{n}{n_0+n}\bar{y}
\end{aligned}$$

Which matches the required value.

For the variance, we'll use $\hat{\Sigma} = [-\nabla^2 \ln(\pi_{post}(\theta|x))]^{-1}$

$$\begin{aligned}
-\frac{d^2}{d\eta^2}\ln(\pi(\eta|y)) &= -\frac{d^2}{d\eta^2} [-(n+n_0)A(\eta) + (n\bar{y}+n_0y_0)\eta] \\
&= -\frac{d}{d\eta} \left[-(n+n_0)\frac{d}{d\eta}A(\eta) + (n\bar{y}+n_0y_0) \right] \\
&= (n+n_0)\frac{d^2}{d\eta^2}A(\eta)
\end{aligned}$$

Plugging this into the above function for $\hat{\Sigma}$:

$$\hat{\Sigma} = \left[(n+n_0)\frac{d^2}{d\eta^2}A(\eta) \right]^{-1}$$

which is what we needed to show.

b)

$f(y_i|\theta) = \text{Bernoulli}(\theta)$. Find a normal approximation for posterior of $\eta = \ln\left(\frac{\theta}{1-\theta}\right)$

$$\begin{aligned}
f(y_i|\theta) &= \theta^y(1-\theta)^{1-y} = e^{y\ln(\theta)+(1-y)\ln(1-\theta)} \\
&= e^{y(\ln(\theta)-\ln(1-\theta))+\ln(1-\theta)} \\
&= e^{y\eta+\ln(1-\frac{e^\eta}{1+e^\eta})} \\
&= e^{y\eta+\ln(\frac{1}{1+e^\eta})} \\
&= e^{y\eta-\ln(1+e^\eta)}
\end{aligned}$$

In this case, $A(\eta) = \ln(1+e^\eta)$

$$\frac{d}{d\eta}\ln(1+e^\eta) = \frac{e^\eta}{1+e^\eta} = \hat{\theta} = \frac{n_0y_0+n\bar{y}}{n_0+n}$$

$$\begin{aligned}
\frac{d^2}{d\eta^2} \ln(1 + e^\eta) &= \frac{d}{d\eta} e^\eta (1 + e^\eta)^{-1} \\
&= e^\eta (1 + e^\eta)^{-1} - e^\eta e^\eta (1 + e^\eta)^{-2} \\
&= \frac{e^\eta}{1 + e^\eta} - \left(\frac{e^\eta}{1 + e^\eta} \right)^2
\end{aligned}$$

We then have a normal approximation with mean:

$$\hat{\theta} = \frac{n_0 y_0 + n \bar{y}}{n_0 + n},$$

$$\hat{\eta} = \ln \left(\frac{\hat{\theta}}{1 - \hat{\theta}} \right)$$

and variance:

$$\hat{\Sigma} = \left[\frac{e^{\hat{\eta}} (n_0 + n)}{1 + e^{\hat{\eta}}} - (n_0 + n) \left(\frac{e^{\hat{\eta}}}{1 + e^{\hat{\eta}}} \right)^2 \right]^{-1}$$

c)

$$n_0 = 2, y_0 = 0.5, \sum_{i=1}^8 y_i = 5 \implies n = 8, \bar{y} = \frac{5}{8}$$

```

n0 <- 2
y0 <- 0.5
n <- 8
y_bar <- 5/8
sum_y <- 5

theta_hat <- (n0*y0 + n*y_bar)/(n0 + n)
eta_hat <- log(theta_hat/(1-theta_hat))
sigma_hat <- ((n0 + n)*(exp(eta_hat)/(1+exp(eta_hat))) - (exp(eta_hat)/(1+exp(eta_hat)))^2))^(-1)

# Simulate samples
nsim <- 10e3
x <- rnorm(nsim, eta_hat, sqrt(sigma_hat))
w <- exp(-(n0+n)*log(1+exp(x)) + (n0*y0 + sum_y)*x) / dnorm(x, eta_hat, sqrt(sigma_hat))
wn <- w/sum(w)

# P(theta > 0)
p_eta_0 <- sum((x > 0)*wn)

```

Thus, $P(\eta > 0|y) = 0.7402098$.

d)

Use a transformation to show the conjugate prior for η in the Bernoulli model is same as a Beta prior for $\theta = (1 + e^{-\eta})^{-1}$. Identify shape params as functions of n_0, y_0 .

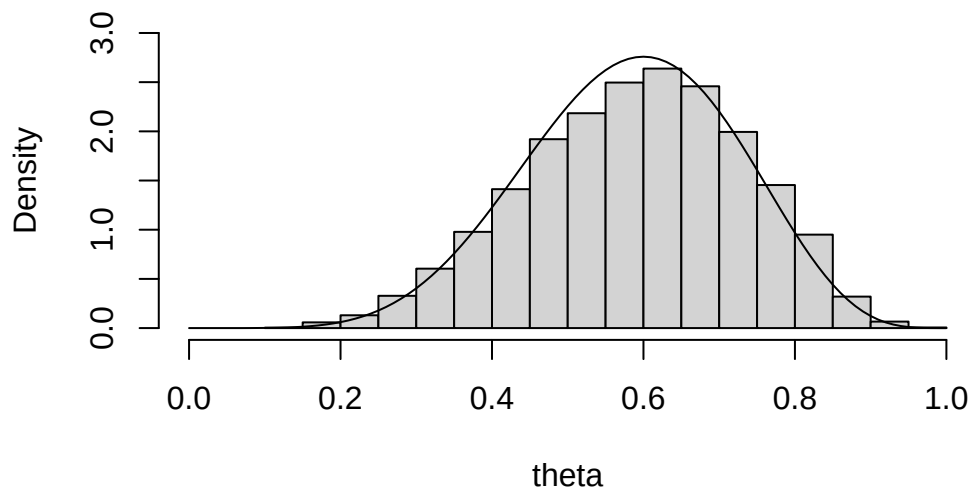
$$\begin{aligned}
 \pi(\eta) &= e^{-n_0 \ln(1+e^\eta) + n_0 y_0 \eta} \implies \\
 \pi(\theta) &= e^{-n_0 \ln(1+e^{\ln(\frac{\theta}{1-\theta})}) + n_0 y_0 \ln(\frac{\theta}{1-\theta})} e^{\ln(\frac{\theta}{1-\theta})} \\
 &= (1 + \frac{\theta}{1-\theta})^{-n_0} \left(\frac{\theta}{1-\theta} \right)^{n_0 y_0} \frac{\theta}{1-\theta} \\
 &= (1-\theta)^{n_0} \theta^{n_0 y_0} (1-\theta)^{-n_0 y_0} \theta (1-\theta)^{-1} \\
 &= \theta^{n_0 y_0 + 1} (1-\theta)^{n_0(1-y_0) + 1} \\
 &= \underline{Beta(n_0 y_0 + 1, n_0(1-y_0) + 1)}
 \end{aligned}$$

e)

$$\pi(\theta|y) \propto \theta^{\sum y_i} (1-\theta)^{n - \sum y_i} \theta^{n_0 y_0} (1-\theta)^{n_0(1-y_0)} = Beta(\sum y_i + n_0 y_0 + 1, n - \sum y_i + n_0(1-y_0) + 1)$$

```
# Transform the previously drawn Etas into Thetas
theta <- 1/(1 + exp(-x))
# Plot densities of sampled thetas and Beta posterior
hist(theta, freq = FALSE, xlim = c(0, 1), ylim = c(0, 3))
curve(dbeta(x, sum_y + n0*y0 + 1, n - sum_y + n0 * (1 - y0) + 1), add = TRUE)
```

Histogram of theta



3

a)

If we take the sum of each row we get: $0 + .4 + .6 = 1, .1 + .3 + .6 = 1, .7 + .2 + .1 = 1$

Thus all rows sum to 1, and this is a valid transition matrix.

b)

To show this is ergodic for MCMC, we need to show that we can get everywhere from anywhere. Basically, we need to show that as n increases, there are no 0s in the transition matrix.

```
p = matrix(c(0, 0.4, 0.6, 0.1, 0.3, 0.6, 0.7, 0.2, 0.1), nrow = 3, byrow = TRUE)
p%%p
```

```
      [,1] [,2] [,3]
[1,] 0.46 0.24 0.30
[2,] 0.45 0.25 0.30
[3,] 0.09 0.36 0.55
```

Going just one step in the future we lose the '0' in the transition matrix, so yes, this is a valid transition matrix.

c)

```
p%%p
```

```
      [,1] [,2] [,3]
[1,] 0.46 0.24 0.30
[2,] 0.45 0.25 0.30
[3,] 0.09 0.36 0.55
```

$P(Y_3 = 3 | Y_1 = 1) = .3$ (the top right cell going from (1) to (3))

d)

```
simulate <- function(n_sim, final_step, starting_pos) {
  res <- vector(mode = 'list', length = n_sim)
  for (i in 1:n_sim) {
    pos <- starting_pos
    for (j in 1:(final_step-1)) {
      pos <- sample(c(1, 2, 3), size = 1, prob = p[pos, ])
    }
    res[[i]] <- pos
  }
  res
}

p_y_3 <- mean(simulate(n_sim = 10e3, final_step = 3, starting_pos = 1) == 3)
```

By simulation, $P(Y_3 = 3 | Y_1 = 1) \approx 0.2994$

e)

```
mult_p <- function(n) {  
  p_next <- p  
  for (i in 1:(n-1)) {  
    p_next <- p_next%*%p  
  }  
  p_next  
}  
  
mult_p(n = 12)
```

	[,1]	[,2]	[,3]
[1,]	0.3092374	0.2908603	0.3999023
[2,]	0.3092374	0.2908603	0.3999023
[3,]	0.3088712	0.2909823	0.4001465

Per the matrix, $P(Y_{12} = 1|Y_1 = 2) = .309$ (left-middle cell: starting at (2), going to (1)).

f)

```
p_y_12 <- mean(simulate(n_sim = 10e3, final_step = 12, starting_pos = 2) == 3)
```

By simulation, $P(Y_{12} = 1|Y_1 = 2) \approx 0.4027$

g)

P^5 :

```
mult_p(n = 5)
```

	[,1]	[,2]	[,3]
[1,]	0.29034	0.29716	0.41250
[2,]	0.29035	0.29715	0.41250
[3,]	0.33721	0.28154	0.38125

P^{10} :

```
mult_p(n = 10)
```

	[,1]	[,2]	[,3]
[1,]	0.3096768	0.2907138	0.3996094
[2,]	0.3096768	0.2907138	0.3996094
[3,]	0.3082120	0.2912021	0.4005859

P^{100} :

```
mult_p(n = 100)
```

```
      [,1]      [,2] [,3]  
[1,] 0.3090909 0.2909091 0.4  
[2,] 0.3090909 0.2909091 0.4  
[3,] 0.3090909 0.2909091 0.4
```

h)

```
mult_p(n = 10000)
```

```
      [,1]      [,2] [,3]  
[1,] 0.3090909 0.2909091 0.4  
[2,] 0.3090909 0.2909091 0.4  
[3,] 0.3090909 0.2909091 0.4
```

The distribution as $n \rightarrow \infty$ is $P(1) = .309$, $P(2) = .2909$, $P(3) = .4$

i)

```
n_sim <- 10000  
steps <- vector(mode = 'list', length = n_sim)  
steps[[1]] <- 1  
for (i in 2:(n_sim)) {  
  steps[[i]] <- sample(c(1, 2, 3), size = 1, prob = p[steps[[i-1]], ])  
}  
p1 <- mean(steps == 1)  
p2 <- mean(steps == 2)  
p3 <- mean(steps == 3)
```

By simulation: the longer term probabilities are: $P(1) \approx 0.3047$, $P(2) \approx 0.2954$, $P(3) \approx 0.3999$

This appears to be the same as the calculated limit in part (h)

Acknowledgements

Estimated time

I estimate this assignment took me in total around 5 hours to complete

Resources Used

I used some googling to try to understand what Question 2 was asking about in some of the steps (specifically to clarify the different transformations I was supposed to do), as well as class code/slides.